

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.datasets import mnist, cifar10

(mnist_x, _), _ = mnist.load_data()
(cifar_x, _), _ = cifar10.load_data()

pixel_a = 50
pixel_b = 150

difference = abs(pixel_b - pixel_a)

print("Task 1: Computation")
print("Pixel A =", pixel_a)
print("Pixel B =", pixel_b)
print("Intensity Difference =", difference)

if difference >= 50:
    print("Strong edge")
else:
    print("Weak edge")

gray_images = []

for i in range(5):
    gray = np.mean(cifar_x[i], axis=2)
    gray_images.append(gray)

edge_images = []

for img in gray_images:
    gx = np.abs(np.diff(img, axis=1))
    gy = np.abs(np.diff(img, axis=0))

    gx = np.pad(gx, ((0,0),(0,1)))
    gy = np.pad(gy, ((0,1),(0,0)))

    edges = np.sqrt(gx**2 + gy**2)
    edges = (edges > 40) * 255

    edge_images.append(edges)

plt.figure(figsize=(15,6))

for i in range(5):
    plt.subplot(2,5,i+1)
    plt.imshow(cifar_x[i])
    plt.title("CIFAR Original")
    plt.axis("off")

for i in range(5):
    plt.subplot(2,5,i+6)
    plt.imshow(edge_images[i], cmap="gray")
    plt.title("Edges")
    plt.axis("off")
```

```
plt.tight_layout()  
plt.show()
```

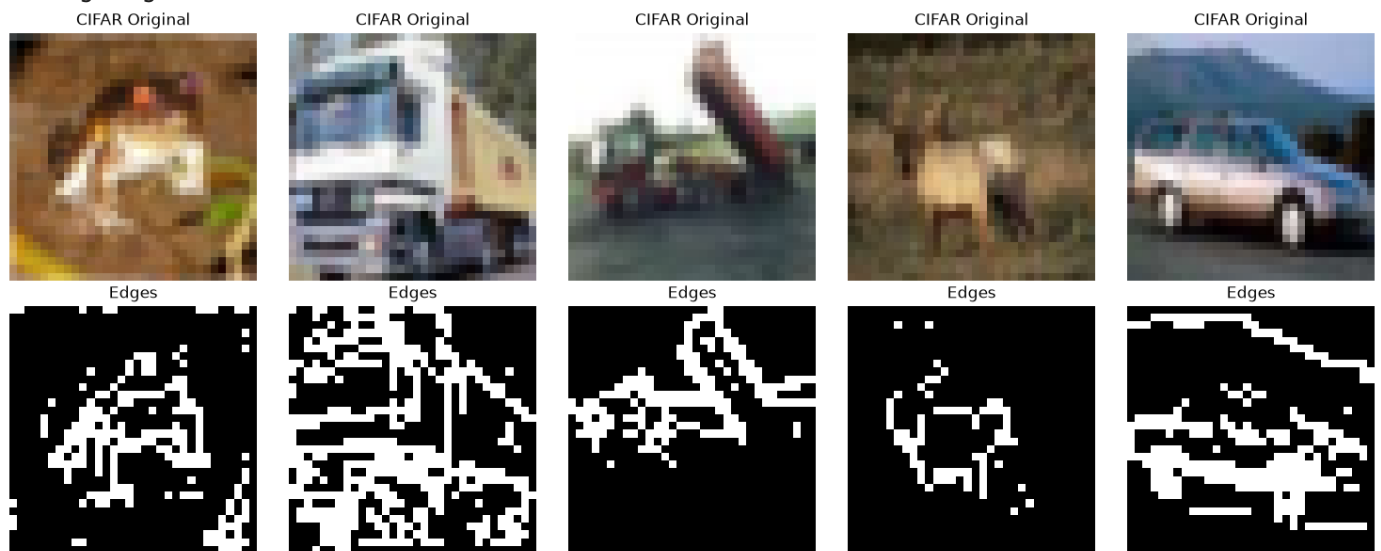
Task 1: Computation

Pixel A = 50

Pixel B = 150

Intensity Difference = 100

Strong edge



In []: